

VolGAN Project Correction Guide

What to correct in the CS 787 Group 26 proposal, how to correct it, and what happens if you do not

1. Purpose and how to read this guide

This guide compares your project proposal (Volatility Surface Arbitrage terminal, four modules) with its base paper, *VolGAN: A Generative Model for Arbitrage-Free Implied Volatility Surfaces* (Vuletić and Cont, Applied Mathematical Finance, 31(4), 2024). For every problem it gives four things: **what is wrong**, **how to correct it**, **what happens if you do not**, and **where the evidence is**.

The pitch itself is worth pursuing. Combining anomaly detection, a generative stress-test and an explanation layer is a coherent idea, and your existing Carr-Madan / Variance Gamma pricer is a real advantage. The problems below are mostly places where the proposal describes the paper inaccurately or leaves a handoff undefined. They are cheap to fix now and expensive to discover during evaluation.

Two kinds of finding

- **PAPER** means the paper says something different from the proposal, or contains a result that bears on it. The page, equation or table is cited so you can check it yourself.
- **JUDGMENT** means it is my engineering assessment, not something the paper states. These can be argued with, and each says how you can test it.

Severity

CRITICAL	Must fix. It makes a claim false, a module untestable, or the output meaningless.
IMPORTANT	Should fix. It weakens the result or invites an obvious challenge.
POLISH	Polish. Small effect, cheap to fix.

Scope note. This is an audit of the proposal text against the paper. No code was reviewed. Where a claim depends on your data (for example how large the Variance Gamma residuals are on NSE quotes), the guide says how to test it instead of asserting the answer.

2. Executive summary

27 issues found: 10 critical, 15 important, 2 polish. Three findings explain most of them.

- **The proposal describes VolGAN differently from the paper in four places.** The paper uses BCE plus smoothness penalties (not Wasserstein-GP), does not put the arbitrage penalty in the loss, outputs a joint (return, surface increment) pair, and measures arbitrage with three terms rather than one (G1, 3.1, 3.2, 3.3).
- **Module 1 mixes units and mostly measures model error.** The FFT gives prices while the market matrix holds implied vols, and a single Variance Gamma fit leaves structured residuals. There are also no training labels (1.1, 1.2, 1.3).
- **The handoff from Module 3 to Module 4 is undefined.** Nobody can say what the trade is or what "survived" means, and the paper's own quantile tables warn that simulated tails can be too thin for a stress-test (4.3, 3.4).

Fix these first, in this order

#	Issue	Action
1	G1	Correct the description of VolGAN and turn the differences into stated extensions with ablations.
2	1.1	Make the Module 1 residual unit-consistent (FFT price to Black-Scholes IV).
3	1.2	Calibrate per expiry with vega weights, and use a leave-out residual.
4	1.3	Create labels (synthetic injection and/or reversion) and split by time.
5	3.1	Generate the joint ($\mathbb{R}$, $\Delta\hat{g}$) output.
6	3.2	Add the Sobolev smoothness penalties with gradient-norm matching.
7	3.3	Use all three arbitrage terms.
8	4.3	Define the trade and the meaning of "survive".
9	4.1	Move the decision to a deterministic rule; the LLM explains only.
10	3.4	Backtest tail exceedance before trusting any survival ratio.

3. Issue register

One line per issue. The detailed correction for each ID is in Section 4.

ID	Severity	Module	Mistake	Basis
G1	CRITICAL	Global	The proposal says VolGAN "strictly enforces" no-arbitrage inside the loss	Paper
G2	IMPORTANT	Global	"Arbitrage" is used for two different things	Judgment
1.1	CRITICAL	Module 1	Units mismatch: FFT gives prices, the market matrix holds implied vols	Judgment
1.2	CRITICAL	Module 1	The residual mostly measures VG model misfit, not market mispricing	Judgment
1.3	CRITICAL	Module 1	No ground-truth labels for the CNN-LSTM	Judgment
1.4	IMPORTANT	Module 1	Output (scalar + coordinate) does not match the promised heatmap	Judgment
1.5	IMPORTANT	Module 1	The LSTM has no defined temporal input	Judgment
1.6	IMPORTANT	Module 1	Raw strikes on a moving underlying, and no handling of missing quotes	Judgment
1.7	IMPORTANT	Module 1	No liquidity or transaction-cost filter	Judgment
1.8	IMPORTANT	Module 1	No simple baseline to justify the CNN-LSTM-Attention stack	Judgment
1.9	POLISH	Module 1	Claim that Module 3 is "conditioned on verified anomalies"	Judgment
2.1	IMPORTANT	Module 2	PCA is dismissed as "linear only" without testing it	Paper
2.2	IMPORTANT	Module 2	"Macroeconomic regime" is an overclaim for a latent built from surfaces only	Judgment
2.3	IMPORTANT	Module 2	Risk of posterior collapse, so c carries no information	Judgment
3.1	CRITICAL	Module 3	The generator outputs surfaces only; the paper outputs (return, surface increment)	Paper

ID	Severity	Module	Mistake	Basis
3.2	CRITICAL	Module 3	Smoothness penalties L_m and L_t are missing	Paper
3.3	CRITICAL	Module 3	L_{arbi} covers calendar arbitrage only	Paper
3.4	CRITICAL	Module 3	Stress realism: the paper's own quantile tests show forecasts that are too narrow	Paper
3.5	IMPORTANT	Module 3	Switch to Wasserstein-GP is untested and changes several other pieces	Paper
3.6	IMPORTANT	Module 3	"Mathematically possible futures" is contradicted by the paper's own numbers	Paper
3.7	IMPORTANT	Module 3	Training surfaces are smoothed, which erases what Module 1 must detect	Paper
3.8	POLISH	Module 3	Moneyness K/S ignores the forward	Judgment
3.9	IMPORTANT	Module 3	Network size and balance are not specified	Paper
4.1	CRITICAL	Module 4	The LLM makes the execute/abort decision	Judgment
4.2	IMPORTANT	Module 4	A single scalar gives the LLM nothing to reason about	Judgment
4.3	CRITICAL	Module 4	"Survived" is never defined	Judgment
E1	IMPORTANT	Evaluation	No evaluation plan	Paper

4. Detailed corrections

Each card lists where the problem is, what is wrong, the evidence, the correction, and the impact of leaving it as it is. Worked procedures referred to as Lessons A to D are in Section 5.

Global claims

G1 · CRITICAL	The proposal says VolGAN "strictly enforces" no-arbitrage inside the loss
Where	Module 3, "Why this over competitors" and the module summary.
Mistake	The proposal claims VolGAN "strictly enforces mathematical physics into the gradient descent loss, ensuring valid generation".
Evidence (paper)	Paper §3.2 (p. 210): the authors say the arbitrage penalty <i>could</i> be added to the generator loss, but they did not, and that trying it made no notable difference. §3.3 (p. 210): outputs "are not guaranteed to satisfy the static arbitrage constraints", so arbitrage is handled afterwards by re-weighting scenarios. Table 1 (p. 216): raw VolGAN has mean arbitrage penalty 0.0199 versus 0.0096 for market data, so it is close to the data but not zero.
How to correct	1. Replace the sentence with an accurate one: "VolGAN encourages smooth, nearly arbitrage-free surfaces through Sobolev smoothness penalties and can re-weight scenarios by $\exp(-\beta\Pi)$. We extend it with an explicit arbitrage penalty and measure whether it helps." 2. Turn the extension into a real experiment: train (a) smoothness only, (b) smoothness + Π in the loss, (c) each of these with re-weighting. Report Π and hedging tracking error for all.
Impact if not corrected	A factual contradiction of your own base paper that anyone who opens page 210 can find. It damages trust in every other claim in the proposal, and it means the phrase "mathematically possible futures" is stated without evidence.
G2 · IMPORTANT	"Arbitrage" is used for two different things
Where	Project title ("Volatility Surface Arbitrage terminal"), Module 1 and Module 3.
Mistake	In the paper, arbitrage means a <i>static no-arbitrage violation inside one surface</i> (for example a negative butterfly price). In Module 1 the signal is a <i>gap from your Variance Gamma fair value</i> , which is a relative-value view, and can just as easily be model error.
Basis (judgment)	Paper §2.1 (pp. 205-206) defines arbitrage through the penalty $\Pi = p_1 + p_2 + p_3$. Nothing in the paper links this to a model-based fair value.
How to correct	1. Rename Module 1 output to "mispricing signal" or "relative-value anomaly". 2. Keep the word "arbitrage" for Π only. 3. In the dashboard show two separate numbers: the mispricing score (Module 1) and the arbitrage penalty Π of the simulated surfaces (Module 3).
Impact if not corrected	Confusion in the viva, and an implied promise of risk-free profit that no part of the pipeline can support. An examiner will ask "where is the riskless portfolio?".

Module 1: Discriminative anomaly detection

1.1 · CRITICAL	Units mismatch: FFT gives prices, the market matrix holds implied vols
Where	Module 1, "Technique & Math".
Mistake	Carr-Madan FFT outputs option prices . The market matrix R contains implied volatilities . Subtracting one from the other is not a meaningful operation.
Basis (judgment)	Carr-Madan is a pricing method (characteristic function to prices). Implied volatility is defined by inverting Black-Scholes (paper §2, p. 205).
How to correct	Follow Lesson A (Section 5): price with FFT, invert Black-Scholes cell by cell to obtain model implied vols, then subtract IV from IV. Use Newton with vega, and fall back to a bracketing method when vega is tiny (deep OTM, very short maturity).
Impact if not corrected	The residual matrix is numerically meaningless. Even if you subtract in price space instead, the residual scales with vega, so ATM cells look huge and wing cells invisible, and the CNN learns the moneyness pattern instead of mispricing.
1.2 · CRITICAL	The residual mostly measures VG model misfit, not market mispricing
Where	Module 1, residual matrix definition.
Mistake	One Variance Gamma parameter set cannot reproduce a whole market surface across all strikes and expiries. Lévy-type models generally flatten the smile with maturity faster than real markets do, so a single global fit leaves large, structured residuals. If instead you calibrate to the same surface you then scan, the fit partly absorbs the anomalies you are trying to find.
Basis (judgment)	Engineering judgment based on standard properties of Lévy models. It is testable: plot residuals from your existing FFT/VG project on a few historical days and check whether they form smooth, repeatable patterns.
How to correct	1. Calibrate per expiry (one VG parameter set for each maturity slice), which is cheap with FFT. 2. Use a weighted objective: minimize $\sum w_k (\sigma_{mkt,k} - \sigma_{VG,k})^2$ with w_k proportional to vega or liquidity. 3. Use a leave-out residual: to score a cell, refit without a small neighbourhood around it, then compare. 4. Report the residual's typical size on "quiet" days as the noise floor.
Impact if not corrected	The CNN detects VG's systematic shape error and reports false anomalies at the same wings and expiries every day. Every downstream stress-test is then run on a trade that was never mispriced.
1.3 · CRITICAL	No ground-truth labels for the CNN-LSTM
Where	Module 1, training.
Mistake	A supervised detector needs labels. Defining "anomaly = large residual" is circular, because the network then only reproduces your threshold.
Basis (judgment)	Not addressed in the proposal.
How to correct	Use one of two label sources: (a) Synthetic injection. Take clean historical residual matrices, add bumps of random location and size, and label them. You now know the truth, and you can measure precision and recall honestly. (b) Reversion label. Mark a cell anomalous if its residual shrinks by at least x% within h days. This ties the label to tradability. Always split by time (train, validation, test in chronological order), never randomly.
Impact if not corrected	No honest accuracy can be reported. Random splits leak future information, so the reported performance would be inflated and would not survive a live test.

1.4 · IMPORTANT	Output (scalar + coordinate) does not match the promised heatmap
Where	Module 1 output, and Final Deliverable.
Mistake	The dashboard promises a 2D heatmap of the detected anomaly, but the module outputs a scalar and a coordinate.
Basis (judgment)	Proposal, Sections 2 and 3.
How to correct	Make the CNN fully convolutional so it produces a per-cell score map. The heatmap is that map; the confidence is its maximum; the coordinate is its argmax. The LSTM and attention then operate on per-day feature maps.
Impact if not corrected	You would have to fabricate the heatmap after the fact with a saliency method, which is less faithful to what the model actually used, and the demo would not match the described model.

1.5 · IMPORTANT	The LSTM has no defined temporal input
Where	Module 1, input format.
Mistake	The input is described as a single matrix, but an LSTM needs a sequence.
Basis (judgment)	Proposal says the LSTM tracks "historical trends".
How to correct	Feed a sequence of L residual matrices (for example $L = 10$ to 20 days). A CNN encodes each day into a vector, the LSTM runs across the days, and attention weights the days.
Impact if not corrected	The LSTM degenerates into an ordinary dense layer, and the stated reason for using it (vanishing gradients over history) becomes untrue.

1.6 · IMPORTANT	Raw strikes on a moving underlying, and no handling of missing quotes
Where	Module 1 input format.
Mistake	Rows are indexed by absolute strike. As spot moves, the same strike sits at a different moneyness each day, and far strikes are often unquoted.
Basis (judgment)	The paper uses a fixed grid in moneyness $m = K/S$ and time-to-maturity τ (§4.1, p. 212) with 11 moneyness levels and 9 maturities.
How to correct	1. Convert to moneyness on a fixed grid. 2. Interpolate onto the grid (the paper interpolates linearly in moneyness, then in maturity). 3. Add a mask channel marking cells with no real quote, so missing data is not read as zero residual.
Impact if not corrected	The CNN sees a different geometry every day and treats empty cells as information, producing fake anomalies on illiquid wings.

1.7 · IMPORTANT	No liquidity or transaction-cost filter
Where	Module 1 output handoff.
Mistake	A residual can be larger than what you could ever capture after spreads.
Basis (judgment)	Not addressed in the proposal.
How to correct	Flag a cell only if $ \text{residual} $ exceeds half the bid-ask width in IV plus a cost buffer. If your data source has only closing or settlement prices and no bid-ask, use volume and open-interest thresholds instead and say so explicitly.
Impact if not corrected	"Anomalies" turn out to be stale or illiquid quotes that cannot be traded, so Module 3 stress-tests a trade that does not exist at that price.

1.8 · IMPORTANT	No simple baseline to justify the CNN-LSTM-Attention stack
Where	Module 1, "Why this over competitors".
Mistake	The argument rejects MLPs but compares against no simple alternative.
Basis (judgment)	Not addressed in the proposal.
How to correct	Add a baseline: a rolling z-score of each residual cell, thresholded. Report precision, recall and (if possible) PnL for baseline versus CNN-LSTM. Claim the CNN only where it beats the baseline.
Impact if not corrected	You will be asked "why not a z-score?" and have no evidence. It is possible the simple rule performs the same.

1.9 · POLISH	Claim that Module 3 is "conditioned on verified anomalies"
Where	Module 1, "Gen AI Involvement".
Mistake	Module 3's input is only the noise z and the VAE vector c . Nothing from Module 1 enters the generator.
Basis (judgment)	Proposal, Module 3 "Input Format".
How to correct	Either reword ("Module 1 chooses <i>which trade</i> to stress-test; Module 3 is conditioned on the market regime"), or add the anomaly location and score as extra conditioning inputs.
Impact if not corrected	A design claim that the architecture does not support.

Module 2: Dimensionality reduction (VAE)

2.1 · IMPORTANT	PCA is dismissed as "linear only" without testing it
Where	Module 2, "Why this over competitors".
Mistake	The proposal rejects PCA outright. The paper itself relies on PCA and shows it captures most of the variance.
Evidence (paper)	Table 4 (p. 224): the first three principal components explain 51.25%, 34.00% and 5.01% of variance of daily log-IV increments in the test data, about 90% in total. §2.2 (p. 206) lists level and skew principal components as standard empirical facts.
How to correct	Keep PCA as a baseline with the same latent size d as the VAE. Compare reconstruction error and, more importantly, the downstream survival ratio and hedging error. Claim the VAE only if it wins.
Impact if not corrected	An untested claim against a strong baseline. The VAE may bring no benefit for extra complexity.

2.2 · IMPORTANT	"Macroeconomic regime" is an overclaim for a latent built from surfaces only
Where	Module 2, output description.
Mistake	The VAE sees only implied-vol surfaces, not macroeconomic variables.
Basis (judgment)	Proposal, Module 2.
How to correct	Call it a "market-state latent". Validate it: plot the latent over time, check that known stress episodes (for example March 2020) separate from calm periods, and correlate it with the India VIX level.
Impact if not corrected	An easily challenged claim with no supporting evidence.

2.3 · IMPORTANT	Risk of posterior collapse, so c carries no information
Where	Module 2 training.
Mistake	With limited data and a powerful decoder, a VAE can ignore its latent and reconstruct from the decoder alone.
Basis (judgment)	Standard VAE behaviour. The paper's PCA results suggest about three effective factors, so a small latent is enough.
How to correct	Use a small d (about 3 to 8). Use KL warm-up or a β -VAE. Monitor the KL of each latent dimension. Run a sensitivity test : change c and check that Module 3's outputs change.
Impact if not corrected	Module 3 silently becomes unconditional, and the claim of regime-conditioned stress-testing is empty.

Module 3: Generative stress-testing (VolGAN)

3.1 · CRITICAL	The generator outputs surfaces only; the paper outputs (return, surface increment)
Where	Module 3, "Output & Handoff".
Mistake	Your generator produces $N \times M$ matrices. The paper's generator produces the pair ($R_t, \Delta \hat{g}_t$): the underlying's return and the log-IV increment, jointly.
Evidence (paper)	Eq. (10) and Figure 1 (p. 207-208). The joint output is what reproduces the leverage effect: Table 6 (p. 227) shows correlation between returns and the level-projection of -0.76 in data and -0.84 in VolGAN.
How to correct	Output both R and $\Delta \hat{g}$ (Lesson C). Then the new spot is $S \cdot \exp(R)$ and the new surface is $\exp(g + \Delta \hat{g})$, so spot and surface move together in each scenario.
Impact if not corrected	You cannot compute delta, hedge, or reprice the trade at the new spot, so Module 4 has nothing to evaluate. The negative spot-vol relationship is lost.

3.2 · CRITICAL	Smoothness penalties L_m and L_τ are missing
Where	Module 3, "Technique & Math".
Mistake	The paper says a plain BCE-trained GAN produces irregular surfaces, and adds discrete Sobolev penalties in moneyness and maturity. The proposal has neither.
Evidence (paper)	Eqs. (11)-(13) (pp. 208-209); Figure 6 (p. 215). Table 1 (p. 216): mean arbitrage penalty is 2.4635 for the BCE-only GAN versus 0.0199 for VolGAN, roughly 120 times worse.
How to correct	Add L_m and L_τ to the generator loss (Lesson B). Choose α_m, α_τ by gradient-norm matching over a 25-epoch pilot run, as in §3.4 (p. 211).
Impact if not corrected	Jagged surfaces that violate arbitrage conditions by two orders of magnitude more than the paper's model. The survival ratio is then computed on unrealistic surfaces.

3.3 · CRITICAL	L_{arbi} covers calendar arbitrage only
Where	Module 3, custom loss.
Mistake	Only $\partial w / \partial \tau < 0$ (calendar) is penalized. The paper's penalty has three parts.
Evidence (paper)	Eqs. (2)-(5) (p. 206): calendar p_1 , call-price monotonicity p_2 and butterfly (convexity) p_3 , all computed on relative call prices obtained from the surface via Black-Scholes.
How to correct	Implement all three as ReLU of finite differences of relative call prices (Lesson B). Compute them on $\hat{o} = \exp(g + \Delta \hat{g})$, which is automatically positive.
Impact if not corrected	Surfaces with a negative implied density (butterfly violations) pass unchecked. Scenarios containing free-lunch structures inflate the survival ratio.

3.4 · CRITICAL	Stress realism: the paper's own quantile tests show forecasts that are too narrow
Where	Module 3, purpose of the stress-test.
Mistake	A stress-test must reach far into the tails. The paper's quantile tests show the opposite: realized moves fall outside the simulated tails far more often than the nominal 1% and 2.5%.
Evidence (paper)	Table 2 (p. 221), with re-weighting: 25.32% of realized SPX returns fall below the simulated 1% quantile (target 1%) and 83.55% fall below the 99% quantile (target 99%). Table 3 (p. 222), without re-weighting ($\beta = 0$): 4.48% and 93.37%. The paper itself states that VolGAN underestimates extremely high implied-vol returns and VIX (p. 221).
How to correct	Reproduce these exceedance ratios on your data (Lesson D). If tails are too thin: (i) include crisis years in training, (ii) condition the generator on a stressed date's input a_t instead of the current one, (iii) roll the generator out over several days, (iv) as a last resort calibrate a noise scale factor to hit nominal exceedance and disclose it.
Impact if not corrected	"Trade survived 9,800 of 10,000 crashes" gives false comfort, because the simulated crashes are milder than real ones. This is the opposite of what a stress-test is for.

3.5 · IMPORTANT	Switch to Wasserstein-GP is untested and changes several other pieces
Where	Module 3, "Technique & Math".
Mistake	The paper trains with binary cross-entropy (eqs. 13-14). Replacing it with Wasserstein-GP is allowed, but it changes the discriminator's meaning and the gradient scale.
Evidence (paper)	Figure 5 (p. 214): the paper's trained discriminator outputs a probability, and its score below 0.2 at the Covid start is used as an event detector. A Wasserstein critic is unbounded, so that reading no longer applies directly. The paper picks α_m, α_t by matching gradient norms of the BCE term (§3.4), and those values would not transfer.
How to correct	1. Implement the paper's BCE version first as your baseline. 2. Add WGAN-GP as an ablation and redo the α matching for it. 3. If you want an anomaly score from a critic, convert it to a percentile of critic values on the training set.
Impact if not corrected	You cannot say whether your change helped. Mis-scaled α makes smoothness either swamp the adversarial term or be ignored, and small GANs are easy to destabilize.

3.6 · IMPORTANT	"Mathematically possible futures" is contradicted by the paper's own numbers
Where	Module 3, output description.
Mistake	Raw VolGAN samples are close to the data in arbitrage penalty, not arbitrage-free. Re-weighting fixes that on paper but can make results worse.
Evidence (paper)	Table 1 (p. 216): raw 0.0199 versus data 0.0096. §4.2.3 (p. 216): in high-penalty periods a few simulations hold most of the weight, so confidence intervals become very narrow. Figure 23 (p. 231): hedging is better with raw outputs than with re-weighted outputs. The paper also notes (p. 222) that setting $\beta = 0$ can improve forecasts.
How to correct	1. Report the distribution of Π over your 10,000 samples. 2. Report the effective sample size $ESS = 1 / \sum w_i^2$ whenever you re-weight. 3. Make β a dashboard control. Choose it by tracking error and ESS, not by Π alone.
Impact if not corrected	Either the claim is refuted by your own Π numbers, or heavy re-weighting collapses the effective sample and the survival ratio is computed from a handful of scenarios.

3.7 · IMPORTANT	Training surfaces are smoothed, which erases what Module 1 must detect
Where	Data pipeline shared by Modules 1 and 3.
Mistake	The paper trains on kernel-smoothed surfaces. Smoothing removes local bumps, which are exactly the mispricings Module 1 looks for.
Evidence (paper)	§4.1 (pp. 212-213): vega-weighted Nadaraya-Watson smoothing with bandwidths $(h_1, h_2) = (0.002, 0.046)$ chosen to minimize arbitrage penalty on a sample day.
How to correct	Run two data tracks: raw quotes on the fixed grid for Module 1, and smoothed surfaces for Modules 2 and 3. Choose bandwidths on your own data by the same arbitrage-penalty criterion.
Impact if not corrected	If Module 1 runs on smoothed data it finds nothing. If Module 3 trains on raw data it learns quote noise and produces jagged surfaces.

3.8 · POLISH	Moneyness K/S ignores the forward
Where	Grid definition.
Mistake	With interest rates and dividends, the natural centre of the smile is the forward, not spot.
Basis (judgment)	The paper estimates the daily risk-free rate from put-call parity (§4.1, p. 212).
How to correct	Estimate the forward from put-call parity on each day and use $m = K/F$ on the grid.
Impact if not corrected	A small systematic bias in the skew and in the penalties, larger at long maturities.

3.9 · IMPORTANT	Network size and balance are not specified
Where	Module 3, architecture.
Mistake	No sizes are given. The paper deliberately uses very small networks and keeps generator and discriminator balanced.
Evidence (paper)	§3.4 (p. 211): generator has three layers with $H = 16$ and $2H = 32$ neurons and 32-dimensional noise; the discriminator has layers of size 16 and 1. The authors call balance between the two "of the utmost importance". Training uses 10,000 epochs, RMSProp, learning rate 0.0001, batch size 100, one discriminator update per generator update.
How to correct	Start from the paper's sizes and hyperparameters. Enlarge only if validation metrics demand it. The authors' code is public (github.com/milenavuletic/VoIGAN , cited on p. 236) and is a sensible starting point.
Impact if not corrected	With only a few thousand daily observations, a large generator overfits or destabilizes training, and you lose the ability to compare to the paper.

Module 4: Explainable AI (LLM)

4.1 · CRITICAL	The LLM makes the execute/abort decision
Where	Module 4, "Output & Handoff".
Mistake	A stochastic text generator is given a single number and asked to decide whether to trade.
Basis (judgment)	Not addressed in the proposal.
How to correct	Decide with a deterministic rule (for example survival ratio at least 85% and CVaR within a stated limit). The LLM only explains that decision. Validate its text: every number it quotes must appear in the input JSON.
Impact if not corrected	Decisions are not reproducible, can be hallucinated, and cannot be audited. Trade decisions from a language model without a rule are hard to defend.

4.2 · IMPORTANT	A single scalar gives the LLM nothing to reason about
Where	Module 4, "Why this over competitors".
Mistake	The proposal argues LLMs handle "complex multi-variable risk reasoning" but the input is one ratio.
Basis (judgment)	Proposal, Module 4 "Input Format".
How to correct	Pass a structured JSON: PnL quantiles, worst scenario, dominant factor, Π statistics, effective sample size, Module 1 score and location, discriminator or critic percentile.
Impact if not corrected	The LLM adds nothing over an if/else statement, which weakens the justification for the module.

4.3 · CRITICAL	"Survived" is never defined
Where	Handoff from Module 3 to Module 4.
Mistake	The proposal says "trade survived 8,500 / 10,000 crashes" but never says what the trade is or what surviving means.
Basis (judgment)	Paper §5 (pp. 229-236) defines hedged PnL $Z_t = V_t - \hat{V}_t$, tracking error, and VaR, which you can reuse.
How to correct	Define the trade (for example the anomalous option, delta-hedged) and a survival rule (Lesson C): scenario j survives if hedged $PnL_j \geq -L$, where L is a stated loss limit. Also report 5% VaR and CVaR, and how the ratio changes with L .
Impact if not corrected	Modules 3 and 4 cannot be tested against each other. The headline output of the whole system has no definition.

Evaluation

E1 · IMPORTANT	No evaluation plan
Where	Whole project.
Mistake	The proposal lists deliverables but no metrics for showing the generator is realistic.
Evidence (paper)	The paper provides a ready-made suite: Table 1 (arbitrage), Tables 4-5 (PCA variance and eigenvector inner products), Tables 6-9 (correlations), Tables 2-3 (quantile exceedance), Table 11 (tracking error, VaR).
How to correct	Adopt the checklist in Section 7. Report it for the paper-faithful baseline and for each extension.
Impact if not corrected	Nothing demonstrates that the synthetic surfaces are realistic, so the dashboard is a demo without evidence.

5. Worked corrections (Lessons A to D)

These are the procedures the cards refer to. Symbols follow the paper: $m = K/S$ is moneyness, τ is time to maturity, $\sigma_t(m, \tau)$ is the implied volatility surface, and $g_t = \log \sigma_t$.

Lesson A. A valid Module 1 residual (fixes 1.1, 1.2, 1.6, 1.7)

Idea. Compare like with like. Black-Scholes is only the unit converter that turns any model's prices into implied vols, so the market and the model live on the same scale.

Steps

- For each expiry T_j calibrate the Variance Gamma parameters (σ, ν, θ) to liquid quotes by weighted least squares in IV, with weights proportional to vega.
- Price all strikes of that expiry with Carr-Madan FFT and interpolate to your grid strikes.
- Invert Black-Scholes for each cell with Newton's method, and fall back to a bracketing solver when vega is tiny:

$$\sigma_{n+1} = \sigma_n - (C_{BS}(\sigma_n) - C_{VG}) / \kappa(\sigma_n), \quad \kappa = S \cdot \phi(d_1) \cdot \sqrt{\tau} \text{ (vega; } \phi = \text{standard normal density)}$$

- Residual: $r_{ij} = \sigma_{mkt}(i, j) - \sigma_{VG}(i, j)$. Cells with no real quote get $r = 0$ and a mask value of 0 in an extra channel.
- Leave-out score: to score cell (i, j) , recalibrate without a small neighbourhood of that cell and recompute its residual.
- Flag a cell only if $|r_{ij}| > \frac{1}{2}(\sigma_{ask} - \sigma_{bid}) + c_{cost}$. Without bid-ask data, use volume and open-interest thresholds.

Unit test that catches most bugs. Feed the FFT the Black-Scholes characteristic function. The inversion must return the input σ to about $1e-6$. Variance Gamma with $\theta = 0$ and $\nu \rightarrow 0$ reduces to Black-Scholes with volatility σ , so it should also agree in that limit.

Lesson B. The corrected generator loss (fixes 3.2, 3.3, 3.5, 3.6, 3.9)

Relative call prices. The arbitrage terms are computed on prices obtained from a surface with the Black-Scholes formula (paper eq. 1):

$$c(m, \tau) = N(d_1) - m \cdot e^{-r\tau} \cdot N(d_2), \quad d_1 = [-\ln m + \tau(r + \sigma^2/2)] / (\sigma\sqrt{\tau}), \quad d_2 = d_1 - \sigma\sqrt{\tau}$$

Arbitrage penalty (eqs. 2-5). Each term is a sum of positive parts of finite differences on the grid:

$$\begin{aligned} \Pi &= p_1 + p_2 + p_3 \\ p_1 \text{ (calendar): } &\sum [(c(m_i, \tau_j) - c(m_i, \tau_{j+1})) / (\tau_{j+1} - \tau_j)]^+ \\ p_2 \text{ (monotonicity): } &\sum [(c(m_{i+1}, \tau_j) - c(m_i, \tau_j)) / (m_{i+1} - m_i)]^+ \\ p_3 \text{ (butterfly): } &\sum [\text{slope}_{\text{left}} - \text{slope}_{\text{right}}]^+, \text{ slope} = \text{finite difference of } c \text{ in } m \end{aligned}$$

Check the exact index conventions against eqs. (3)-(5) on page 206 when you code them, because the printed layout of those equations is dense.

Smoothness penalties (eqs. 11-12), discrete Sobolev seminorms of the log-IV surface g :

$$\begin{aligned} L_m(g) &= \sum_{i,j} (g(m_{i+1}, \tau_j) - g(m_i, \tau_j))^2 / |m_{i+1} - m_i|^2 \\ L_\tau(g) &= \sum_{i,j} (g(m_i, \tau_{j+1}) - g(m_i, \tau_j))^2 / |\tau_{j+1} - \tau_j|^2 \end{aligned}$$

Generator and discriminator objectives (eqs. 13-14). The generator output is the pair $G(a_t, z_t) = (R_t, \Delta\hat{g}_t)$, and the simulated surface is $\hat{\sigma} = \exp(g_t + \Delta\hat{g}_t)$:

$$\begin{aligned} J^{(G)} &= -\frac{1}{2} E[\log D(a_t, G(a_t, z_t))] + \alpha_m E[L_m(g_t + \Delta\hat{g}_t)] + \alpha_\tau E[L_\tau(g_t + \Delta\hat{g}_t)] + \alpha_\Pi E[\Pi(\hat{\sigma})] \\ &\text{(your extension)} \\ J^{(D)} &= -\frac{1}{2} E[\log D(a_t, (R_t, \Delta g_t))] - \frac{1}{2} E[\log(1 - D(a_t, G(a_t, z_t)))] \\ a_t &= (R_{t-\Delta t}, R_{t-2\Delta t}, \gamma_{t-\Delta t}, g_t(m, \tau)) \quad (\gamma = \text{one-month realized volatility}) \end{aligned}$$

Choosing the weights by gradient-norm matching (§3.4, p. 211):

- Pilot run of 25 epochs with the BCE term only. At every update record the gradient norms of BCE, L_m , L_τ (and Π) with respect to the generator parameters.
- Set $\alpha_m = \text{mean of } \|\nabla \text{BCE}\| / \|\nabla L_m\|$, and likewise for α_τ and α_Π .
- Restart from the **same initialization**. Train 10,000 epochs with RMSProp, learning rate 0.0001, batch size 100, alternating one discriminator and one generator update.
- Draw $N = 10,000$ raw samples. Sizes: generator 16 and 32 neurons with softplus, noise dimension 32; discriminator 16 then 1 (softplus, sigmoid).

Re-weighting (optional). Given penalties Π_i , the scenario weights are $w_i = \exp(-\beta \Pi_i) / \sum_i \exp(-\beta \Pi_i)$ (eq. 17). Always report the effective sample size $\text{ESS} = 1 / \sum w_i^2$. Eq. (20) sets β day by day, but as printed it is written in terms of the weights themselves, which depend on β . Confirm the exact definition in Cont and Vuletić (2023) or in the authors' code before implementing it.

If you keep Wasserstein-GP. Replace the first term by $-E[D(a_t, G)]$ and add the gradient penalty to the critic. The BCE gradient norms used above no longer apply, so repeat the matching procedure for the new adversarial term. Treat it as an ablation against the BCE baseline, not a replacement.

Lesson C. Joint scenarios, hedged PnL and the meaning of "survive" (fixes 3.1, 4.3)

Define the trade first. For example: the option flagged by Module 1 (long if it is cheap, short if it is rich), hedged daily as in paper §5. Then, for each of the 10,000 scenarios j :

- Sample noise z_j . Generate $(R_j, \Delta \hat{g}_j) = G(a_t, z_j)$.
- New spot and surface:

$$S_j = S_t \cdot \exp(R_j), \quad \sigma_j(m, \tau) = \exp(g_t(m, \tau) + \Delta \hat{g}_j(m, \tau))$$

- Reprice the trade at the next date: $V_j = \text{BS}(S_j, K, \tau - \Delta t, \sigma_j(K/S_j, \tau - \Delta t))$, reading σ_j off the grid by linear interpolation in moneyness then maturity, as the paper does. $\Delta V_j = V_j - V_t$.
- Hedge ratios by regression across scenarios (paper eq. 27), on the underlying and, if you wish, a few options chosen by LASSO at inception and then refitted by OLS (§5.5):

$$\Delta V_j = A + \sum_{i \in H} \phi_i \cdot \Delta H_{i,j} + \varepsilon_j$$

- Hedged PnL of scenario j : $\text{PnL}_j = \Delta V_j - \sum_i \phi_i \Delta H_{i,j}$.
- **Survive** if $\text{PnL}_j \geq -L$, with L a stated loss limit (a fixed amount, or a fraction of the premium). Survival ratio = (number of surviving scenarios) / 10,000.
- Also report 5% VaR, CVaR and the survival ratio as a function of L . Use the deterministic rule (for example ratio $\geq 85\%$ and CVaR within limit) to decide EXECUTE or ABORT.
- For a real stress-test, condition the generator on the input a_t from a **stressed historical date** rather than only on today's input.

Lesson D. Tail check before trusting the survival ratio (fix 3.4)

Exceedance ratio. For a quantile level q , the ratio is the fraction of realized values that fall below the simulated q -quantile. A well-calibrated simulator gives a ratio close to q . The paper reports this in Tables 2 and 3 for $q \in \{1\%, 2.5\%, 97.5\%, 99\%\}$.

- Variables: index return, ATM implied vol at several maturities, and the volatility index (India VIX for NSE).
- With re-weighting use the weighted quantile of eq. (19): $x_{(k)}$ with $k = \min\{j : \sum_{i \leq j} w_{(i)} \geq q\}$.
- Test whether hit rates are consistent with q . A Kupiec proportion-of-failures test is the standard tool.
- Compare with the paper's values (SPX return, 1% quantile: 25.32% with re-weighting, 4.48% without) so you know what "as good as the paper" means.

If tails are too thin, try in this order: longer history that includes crises; conditioning on a stressed date's a_t ; a multi-day roll-out; and, as a disclosed last resort, a noise scale factor calibrated so exceedance meets nominal on validation data.

6. Corrected architecture

The same four modules, with the corrections applied. Arrows show data flow. Items marked as ablation or baseline are there so that every extension can be measured against the paper.

```

RAW NSE chain --> grid in m = K/F, tau (+ mask channel for missing quotes)
|
v
[M1] per-expiry VG fit -> Carr-Madan FFT -> Black-Scholes inversion
    residual = IV_mkt - IV_VG (leave-out, L days)
    CNN (per day) -> LSTM -> Attention => score map (heatmap), argmax, confidence
    + liquidity / cost filter + z-score baseline for comparison
    | chosen trade (K, T)
    |
SMOOTHED surfaces --> [M2] VAE (or PCA baseline) --> market-state latent c (small d)
|
v
a_t = (R_{t-1}, R_{t-2}, realized vol, g_t) + c, noise z
|
v
[M3] VolGAN generator --> (R_j, dg_j), j = 1..10,000
    loss: BCE + a_m*L_m + a_tau*L_tau (+ a_Pi*Pi) <- paper baseline / extension
          WGAN-GP variant <- ablation only
    optional re-weighting exp(-beta*Pi); report ESS
    |
    v
reprice trade on (S_j, sigma_j) -> hedge by regression -> PnL_j
survive if PnL_j >= -L -> ratio, VaR, CVaR
|
v
[M4] deterministic rule: EXECUTE / ABORT ; LLM explains from a structured JSON
|
v
DASHBOARD: anomaly heatmap | 3D surfaces (+ Pi) | tracking-error histogram | LLM text

```

7. Evaluation checklist mapped to the paper

Run the full checklist for the paper-faithful baseline first, then for each extension, and present the results as an ablation table.

Test	What to compute	Paper reference	Suggested pass criterion
Arbitrage penalty Π	Mean, std, median of Π on out-of-sample days for market data, BCE-only GAN, raw VolGAN, extension	Table 1 (p. 216)	Same order of magnitude as market data (paper: 0.0096 data, 0.0199 raw)
Smoothness	L_m, L_τ values and sample surface plots against a real surface	Fig. 6 (p. 215)	No visible jaggedness
PCA structure	Variance explained by top 3 PCs; inner products of simulated versus data eigenvectors over 1,000 samples	Tables 4, 5 (pp. 224, 227)	Level and skew inner products near 0.9 (paper: 0.92, 0.92, 0.80)
Correlations	Return versus PC projections; return, level factor, ATM vol, VIX increments; first and last year separately	Tables 6-9 (pp. 227-228)	Correct signs and comparable magnitudes
Distribution shape	Histograms of returns and ATM-vol increments versus Gaussian	Figs. 18, 19 (pp. 223-224)	Asymmetric, heavy tails
Time-varying correlation	Instantaneous return-vol correlation versus the constant-correlation confidence band	Fig. 17 (p. 223)	Outside the band, as in the paper
Quantile exceedance	Exceedance ratios at 1%, 2.5%, 97.5%, 99% for return, ATM vols, VIX; Kupiec test	Tables 2, 3 (pp. 221-222)	Close to nominal; report honestly where not
Event detection	Discriminator or critic-percentile score around known stress dates	Fig. 5 (p. 214)	Score separates stress dates

Test	What to compute	Paper reference	Suggested pass criterion
Hedging	Tracking error mean, std, 5%, 2.5%, 1% VaR versus BS delta and delta-vega	Table 11 (p. 235)	At least competitive with BS delta
Module 1 detection	Precision and recall on synthetic injections; against z-score baseline (new)	Not in paper	CNN-LSTM beats the baseline
Survival sensitivity	Survival ratio versus loss limit L , β and latent c (new)	Not in paper	Stable, sensible monotone behaviour

8. Suggested replacement wording for the proposal

These are drafts you can adapt. They keep your architecture but describe it accurately and turn every difference from the paper into a stated, testable extension.

Module 1: Technique and math

Input is the raw NSE options chain mapped to a fixed grid in forward moneyness and maturity, with a mask for missing quotes. For each expiry we calibrate a Variance Gamma model, price with Carr-Madan FFT, and invert Black-Scholes to obtain model implied volatilities. The residual matrix is market IV minus model IV, computed with a leave-out fit to avoid absorbing the anomaly. A CNN encodes the residual matrix of each day, an LSTM aggregates the last L days, and an attention layer pools them into a per-cell score map. Labels come from synthetic bump injection and from a reversion criterion, with chronological splits. A rolling z-score is the baseline. We call the output a *mispricing signal*, not an arbitrage.

Module 3: Technique and math

We implement VolGAN as published: a conditional GAN with binary cross-entropy loss and Sobolev smoothness penalties whose weights are set by gradient-norm matching. The generator outputs the joint one-day-ahead return and log-implied-vol increment. We extend it in two ways: (i) an explicit arbitrage penalty $\Pi = p_1 + p_2 + p_3$ (calendar, monotonicity, butterfly) in the generator loss, and (ii) a Wasserstein-GP objective. Each extension is compared with the baseline using the paper's tests. Scenarios may be re-weighted by $\exp(-\beta\Pi)$, and we report effective sample size.

Module 4: Output and handoff

For each generated scenario we reprice and hedge the flagged trade by regression, obtaining a hedged PnL. A scenario survives if $\text{PnL} \geq -L$. The survival ratio, VaR and CVaR are computed deterministically, and a stated rule decides EXECUTE or ABORT. The LLM receives these numbers as structured JSON and produces the explanation; its output is checked so that every quoted number matches the input.

One-sentence framing of the whole project

A VolGAN-based stress-testing engine, extended with regime conditioning and explicit arbitrage constraints, that drives an explainable relative-value trading terminal on NSE options.

9. Build order, risks and fallback

Phase	Work	Detail	Done when
0	Reproduce the paper baseline	Run the authors' public code (github.com/milenavuletic/VolGAN) or a faithful re-implementation on the data it was built for, to confirm your pipeline gives Table 1 and PCA results in the right range.	Baseline numbers close to the paper
1	Data pipeline	Forward moneyness grid, mask channel, raw track for Module 1, smoothed track for Modules 2 and 3.	Fixed-shape tensors, low Π after smoothing
2	Module 1	Per-expiry VG, FFT, BS inversion, residual matrices; synthetic labels; z-score baseline; then the CNN-LSTM-Attention.	Precision and recall beat baseline
3	Module 3 extensions	Add Π to the loss and the WGAN-GP variant as ablations against the baseline.	Ablation table filled
4	Module 2	VAE with small d and KL warm-up, plus PCA baseline; sensitivity test on c .	Changing c changes Module 3 output
5	Survival and hedging	Trade definition, regression hedge, survival ratio, VaR, CVaR, tail check (Lesson D).	Exceedance ratios reported
6	Module 4 and dashboard	Deterministic rule, JSON to LLM, number-consistency check; heatmap, 3D surfaces, histogram, text.	End-to-end run on one date
7	Evaluation and write-up	Run the Section 7 checklist for every variant.	Complete results tables

Risks to check early

- **Data.** Confirm the history length and quote quality of your NSE source, and the exercise style of the contracts you use. The paper's data are 2000-2023 SPX quotes from OptionMetrics, so check how quote quality varies across strikes, expiries and years in your source.
- **Sample size.** A GAN trained on a few thousand daily surfaces needs small networks. Start with the paper's sizes.
- **Calibration cost.** Per-expiry VG calibration with leave-out repeated for every cell can be slow. Restrict the leave-out to flagged candidates.
- **Notation clash.** The paper's $c(m, \tau)$ is a relative call price, while your proposal uses c for the VAE latent. Rename the latent (for example ζ) to avoid confusion in the report.

Fallback tiers

Your proposal promises to consult the professor and get prior approval before adjusting any module. If compute or data forces a reduction, these are the least damaging cuts, in order:

- **Tier B.** Drop the WGAN-GP ablation and use PCA instead of the VAE for regime conditioning. Keep Module 1, the paper-faithful VolGAN, and the survival test.
- **Tier C.** Keep the paper-faithful VolGAN and Module 1 evaluated on synthetic injections, with a rule-based explanation template instead of an LLM. This is the minimum coherent pipeline.

Questions to confirm with the professor

- Is it acceptable to present the Wasserstein-GP and explicit-arbitrage-loss changes as extensions with ablations against the published VolGAN?
- Which NSE data source and history length are acceptable for the project?
- Is the survival definition (hedged PnL above a stated limit) an acceptable measure of "trade survived"?

Appendix A. Notation and paper page map

Symbol	Meaning	Paper
$m = K/S$	Moneyness (use K/F with the forward, see 3.8)	p. 205
$\tau = T - t$	Time to maturity	p. 205
$\sigma_t(m, \tau)$	Implied volatility surface at date t	p. 205
$g_t = \log \sigma_t; \Delta g_t$	Log implied vol surface and its daily increment	p. 207
R_t	Log return of the underlying from t to $t + \Delta t$	p. 207
γ_t	One-month realized volatility	p. 207
a_t	Condition vector (two past returns, realized vol, current surface)	p. 207
G, D	Generator and discriminator networks	pp. 207-209
$c(m, \tau)$	Relative call price (paper). Do not reuse c for the VAE latent.	p. 206
$\Pi = p_1 + p_2 + p_3$	Arbitrage penalty: calendar, monotonicity, butterfly	p. 206
$L_m, L_\tau, \alpha_m, \alpha_\tau$	Smoothness penalties and their weights	pp. 208-211
β, w_i	Re-weighting strength and scenario weights	pp. 210-212
$Z_t = V_t - \hat{V}_t$	Tracking error (PnL of the hedged position)	p. 229

Reference

Vuletić, M. and Cont, R. (2024). VolGAN: A Generative Model for Arbitrage-Free Implied Volatility Surfaces. *Applied Mathematical Finance*, 31(4), 203-238. doi:10.1080/1350486X.2025.2471317.